

COIS3510 Data Visualization Workshop 1 Instructions – Visualizing Univariate Data with R or Python

Configure Your Environment

Please follow these instructions to set up your computer environment:

- [Install R and RStudio on Windows](#)
- [Install R and RStudio on Mac](#)
- [Install R and RStudio on Ubuntu](#)

Alternatively, you may set up a Python-based environment:

- [Installing Python on Windows](#)
- [Installing Python on Mac](#)
- [Installing Matplotlib on Windows](#)
- [Installing Matplotlib](#)
- [Installing Matplotlib on Mac](#)
- [Installing Jupyter on Windows](#)
- [Installing Jupyter on Mac](#)

Complete instructions below either in R, OR in Python.

Univariate Data Plots in R

Open RStudio and complete the following plots by entering code snippets below:

Column Chart

```
# Create a column chart
barchart_data <- c(10, 15, 20, 5)
barplot(barchart_data)
# Add labels
barchart_labels <- c("Mar", "Apr", "May", "Jun")
barplot(barchart_data, names.arg=barchart_labels, xlab="Month", ylab="Sales Revenue", col="red", main="Revenue chart", border="blue")
```

Barchart

```
# Create a barchart by changing the plot orientation
barplot(barchart_data, names.arg=barchart_labels, xlab="Sales Revenue", ylab="Month", col="red", main="Revenue chart", border="blue", horiz=TRUE)
```

Stacked bar chart

```
# Create the input vectors.
colors = c("green","orange","brown")
months <- c("Mar","Apr","May","Jun","Jul")
regions <- c("East","West","North")
# Create the matrix of the values.
values <- matrix(c(2,9,3,11,9,4,8,7,3,12,5,2,8,10,11), nrow = 3, ncol = 5, by
row = TRUE)
# Create the stacked bar chart
barplot(values, main = "Sales Revenue", names.arg = months, xlab = "Revenue",
ylab = "Month", col = colors, horiz=TRUE)
legend("bottomright", regions, cex = 1.3, fill = colors)
```

Pie chart

```
# import ggplot2 and gridExtra
install.packages("ggplot2")
install.packages("gridExtra")
library(ggplot2)
library(gridExtra)
# pie chart
slices <- c(10, 15, 10, 5, 8)
labels <- c("ON","QUE", "SK", "MAN", "BC")
pie(slices, labels = labels, main="Sales by Province")
```

Boxplot

```
# boxplot example
# Generate some normally distributed univariate data
set.seed(1)
distr_data <- rnorm(1000, mean=20.5, sd=5.5)
boxplot(distr_data)
boxplot(distr_data, notch=TRUE)
```

Histogram

```
# histogram example
hist(distr_data)
# change the number of bins
hist(distr_data, breaks = 20)
```

Density Plot

```
d <- density(distr_data)
> plot(d, main="Sale Amount Distribution")
> polygon(d, col="red", border="blue")
```

Beeswarm Plot

```
# install the beeswarm package
install.packages("beeswarm")
library(beeswarm)

# plot with default settings
beeswarm(distr_data)

# change the orientation, colour, and symbol
beeswarm(distr_data, col=2, pch=18, cex=1.5, horizontal=TRUE)
```

Violin Plot

```
# using a built-in dataset
ToothGrowth$dose <- as.factor(ToothGrowth$dose)
head(ToothGrowth)

# basic plot
p <- ggplot(ToothGrowth, aes(x=dose, y=len)) + geom_violin()
p

# add stats
p + geom_boxplot(width=0.1)

# add beehive
p + geom_dotplot(binaxis='y', stackdir='center', dotsize=1)
```

Univariate Data Plots in Python

Start your Jupyter notebook, and complete the following plots by entering code snippets below:

```
import matplotlib.pyplot as plt

import numpy as np

import pandas as pd

import seaborn as sns

from scipy.stats import gaussian_kde

# Data

barchart_data = [10, 15, 20, 5]
```

```
barchart_labels = ["Mar", "Apr", "May", "Jun"]
```

```
# Create a bar chart
```

```
plt.bar(barchart_labels, barchart_data, color='red', edgecolor='blue')
```

```
# Add labels and title
```

```
plt.xlabel("Month")
```

```
plt.ylabel("Sales Revenue")
```

```
plt.title("Revenue chart")
```

```
# Display the chart
```

```
plt.show()
```

```
# Create a horizontal bar chart
```

```
plt.barh(barchart_labels, barchart_data, color='red', edgecolor='blue')
```

```
# Add labels and title
```

```
plt.xlabel("Month")
```

```
plt.ylabel("Sales Revenue")
```

```
plt.title("Revenue chart")
```

```
# Display the chart
```

```
plt.show()
```

```
##### Stacked Bar Chart
```

```
# Create the input vectors
```

```
colors = ["green", "orange", "brown"]
```

```
months = ["Mar", "Apr", "May", "Jun", "Jul"]
```

```
regions = ["East", "West", "North"]
```

```
# Create the matrix of the values
```

```
Values = np.array([  
    [2, 9, 3, 11, 9],  
    [4, 8, 7, 3, 12],  
    [5, 2, 8, 10, 11]  
])
```

```
# Plot the stacked bar chart
```

```
plt.barh(months, Values[0], color=colors[0])  
plt.barh(months, Values[1], left=Values[0], color=colors[1])  
plt.barh(months, Values[2], left=Values[0] + Values[1], color=colors[2])
```

```
# Add labels and title
```

```
plt.xlabel("Revenue")  
plt.ylabel("Month")  
plt.title("Sales Revenue")
```

```
# Add legend
```

```
plt.legend(regions, loc="lower right", fontsize=13)
```

```
# Display the chart
```

```
plt.show()
```

```
##### Pie Chart
```

```
# Data
```

```
slices = [10, 15, 10, 5, 8]
```

```
labels = ["ON", "QUE", "SK", "MAN", "BC"]
```

```
# Create the pie chart
```

```
plt.pie(slices, labels=labels, autopct='%1.1f%%')
```

```
plt.title("Sales by Province")
```

```
plt.show()
```

```
##### Boxplot
```

```
# Generate some normally distributed univariate data
```

```
np.random.seed(1)
```

```
distr_data = np.random.normal(20.5, 5.5, 1000)
```

```
# Basic boxplot
```

```
plt.boxplot(distr_data)
```

```
plt.show()
```

```
# Boxplot with notches
```

```
plt.boxplot(distr_data, notch=True)
```

```
plt.show()
```

```
##### Histogram
```

```
# Basic histogram
```

```
plt.hist(distr_data)
```

```
plt.show()
```

```
# Histogram with a different number of bins
```

```
plt.hist(distr_data, bins=20)
```

```
plt.show()
```

Density Plot

Generate density plot

```
density = gaussian_kde(distr_data)
x = np.linspace(min(distr_data), max(distr_data), 1000)
plt.plot(x, density(x), label="Density")
plt.fill_between(x, density(x), color="red", edgecolor="blue")
plt.title("Sale Amount Distribution")
plt.show()
```

Beeswarm Plot

Beeswarm plot

```
sns.swarmplot(distr_data)
plt.show()
```

Customize orientation, color, and symbol

```
sns.swarmplot(distr_data, color='red', marker='^', size=4, orient='h')
plt.show()
```

Violin Plot

Load the dataset from the GitHub URL

```
url =
"https://raw.githubusercontent.com/vincentarelbundock/Rdatasets/master/csv/datasets/ToothGrowth.csv"
data = pd.read_csv(url)
```

```

# Convert 'dose' to a categorical variable
data['dose'] = data['dose'].astype('category')

# Prepare the data for violin plot
grouped_data = [group['len'].values for _, group in data.groupby('dose')]

# Create the violin plot
plt.violinplot(grouped_data, showmeans=False, showmedians=True)

# Add boxplot
plt.boxplot(grouped_data, positions=[1, 2, 3], widths=0.1)

# Add dotplot (scatter plot for individual points)
for i, group in enumerate(grouped_data, start=1):
    plt.scatter([i] * len(group), group, color='black', s=5, alpha=0.6)

# Customize axis labels
plt.xticks([1, 2, 3], ['0.5', '1.0', '2.0'])
plt.xlabel('Dose')
plt.ylabel('Tooth Length')
plt.title('Violin Plot of Tooth Growth by Dose')
plt.show()

```

Submission Instructions

Please submit your R markdown file, or your Jupyter notebook, AND a PDF showing your generated plots via Blackboard by the deadline indicated in Blackboard. If you are using Python please submit you Jupyter Notebook, and a PDF containing your visualizations.